

# The $\tau_\infty$ density condition need not imply $p$ -OAP

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## Source problem

The source paper is J. A. Chavez-Dominguez, V. Dimant, and D. Galicer, *The  $p$ -Operator Approximation Property*, arXiv:2410.05014 [1]. The question appears on PDF page 12, after Theorem 3.8.

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compactness—such as complete compactness and operator compactness—which all generalize the classical one. It would be interesting to know whether there is an operator space satisfying the density condition in  $\tau_\infty$  but the space does not have  $p$ -OAP.

Similarly to the classical context [SK02, Thm. 6.4] (see also [DOPS09, Cor. 2.5]), the 2-OAP holds in every operator space.

The relevant theorem context is the following. The paper proves a list of equivalent conditions for  $p$ -OAP, and remarks that the topology  $\tau_{cc}$  is used in part (iv) instead of the perhaps expected  $\tau_\infty$ .

Now we present a noncommutative version of (a part of) [DOPS09, Thm. 2.1] and [SK02, Thm. 4.2]. Surprisingly, in part (iv), we need to employ the topology  $\tau_{cc}$  instead of the perhaps expected  $\tau_\infty$  (uniform convergence on operator compact matrix sets). For the case  $p = \infty$ , it should be noted that Webster [Web98, Thm. 4.4] already proved the equivalence (i)  $\Leftrightarrow$  (iii).

**Theorem 3.8.** *Let  $1 \leq p \leq \infty$  and let  $V$  be an operator space. The following are equivalent:*

- (i)  $V$  has  $p$ -OAP.
- (ii) For every operator space  $W$ ,  $\mathcal{F}(W, V)$  is  $\tau_p$ -dense in  $\mathcal{K}_\infty^o(W, V)$ .
- (iii) For every operator space  $W$ ,  $\mathcal{F}(W, V)$  is  $\|\cdot\|_{cb}$ -dense in  $\mathcal{K}_p^o(W, V)$ .
- (iv) For every operator space  $W$ ,  $\mathcal{F}(W, V)$  is  $\tau_{cc}$ -dense in  $\mathcal{K}_p^o(W, V)$ .

*Proof.* (i)  $\Rightarrow$  (ii) is trivial from Proposition 3.5.

(ii)  $\Rightarrow$  (iii): Fix  $T \in \mathcal{K}_p^o(W, V)$ . By Theorem 2.4 there exist an operator space  $Z$ , an operator  $p$ -compact map  $R : W \rightarrow Z$ , and an operator compact map  $S : Z \rightarrow V$  such that  $T = SR$ . By our

Thus the natural reading of the question is:

For a fixed  $p$ , can there be an operator space  $V$  such that, for every operator space  $W$ , the finite-rank maps  $\mathcal{F}(W, V)$  are  $\tau_\infty$ -dense in  $\mathcal{K}_p^o(W, V)$ , but  $V$  does not have  $p$ -OAP?

## Proof Intuition

The topology  $\tau_\infty$  is a compact-open type topology: it tests uniform convergence on operator compact matrix sets. Without a boundedness condition on the approximating finite-rank maps, compact-open density is very weak. On a fixed completely compact matrix set in  $W$ , one first approximates all matrix levels by a finite-dimensional subspace  $E \subset W$ . A finite-rank completely bounded projection  $P : W \rightarrow E$  then makes  $T \circ P$  a finite-rank map that is uniformly close to  $T$  on that matrix set. This works for every completely bounded  $T : W \rightarrow V$ , not just for  $p$ -compact maps.

## A compact-open density lemma

**Lemma.** *Let  $W, V$  be operator spaces. Then  $\mathcal{F}(W, V)$  is  $\tau_{cc}$ -dense in  $\mathcal{CB}(W, V)$ , where  $\tau_{cc}$  is uniform convergence on completely compact matrix sets. Consequently  $\mathcal{F}(W, V)$  is  $\tau_\infty$ -dense in  $\mathcal{CB}(W, V)$ , and hence in every subspace of  $\mathcal{CB}(W, V)$ .*

*Proof.* Fix  $T \in \mathcal{CB}(W, V)$ , a completely compact matrix set  $\mathbf{L} = (L_n)_n$  over  $W$ , and  $\varepsilon > 0$ . By complete compactness, there is a finite-dimensional subspace  $E \subseteq W$  such that every  $x \in L_n$  is within  $\delta$  of some point of  $M_n(E)$ , for all  $n$ , where  $\delta > 0$  will be chosen below.

Choose a basis  $e_1, \dots, e_m$  of  $E$ . Let  $a_1, \dots, a_m \in E^*$  be the coordinate functionals and extend them by Hahn–Banach to  $\varphi_1, \dots, \varphi_m \in W^*$ . The map

$$P(w) = \sum_{j=1}^m \varphi_j(w) e_j$$

is finite-rank and completely bounded, since scalar-valued bounded functionals are completely bounded and finite sums of rank-one completely bounded maps are completely bounded. Also  $P|_E = \text{id}_E$ .

Set  $S = T \circ P$ . Then  $S \in \mathcal{F}(W, V)$ . For  $x \in L_n$ , choose  $e \in M_n(E)$  with  $\|x - e\| < \delta$ . Since  $P_n e = e$ ,

$$\|(T - S)_n x\| = \|T_n(x - P_n x)\| \leq \|T\|_{cb} \|x - P_n x\| \leq \|T\|_{cb} (1 + \|P\|_{cb}) \delta.$$

Choosing  $\delta < \varepsilon / (\|T\|_{cb} (1 + \|P\|_{cb}))$  gives uniform  $\varepsilon$ -approximation on  $\mathbf{L}$ . This proves  $\tau_{cc}$ -density.

Finally, every operator compact matrix set is completely compact [1], so  $\tau_\infty$  is weaker than  $\tau_{cc}$ . Thus  $\tau_{cc}$ -density implies  $\tau_\infty$ -density.  $\square$

## Full answer to the source question

**Theorem.** *For every  $2 < p \leq \infty$ , there exists an operator space  $V$  such that, for every operator space  $W$ ,  $\mathcal{F}(W, V)$  is  $\tau_\infty$ -dense in  $\mathcal{K}_p^o(W, V)$ , while  $V$  fails  $p$ -OAP. For  $1 \leq p \leq 2$ , no such example exists.*

*Proof.* The density condition is automatic by the lemma: for every  $W, V$ ,  $\mathcal{F}(W, V)$  is  $\tau_\infty$ -dense in all of  $\mathcal{CB}(W, V)$ , hence also in  $\mathcal{K}_p^o(W, V)$ .

It remains only to choose  $V$  without  $p$ -OAP. The source paper recalls that for every  $2 < p \leq \infty$ , Sinha–Karn’s theorem gives a Banach space without  $p$ -AP, and therefore, by the source paper’s Proposition 3.2, any compatible operator space structure on that Banach space gives an operator space without  $p$ -OAP.

which implies that  $V$  has the  $p$ -AP [SK02, Comment between Defn. 6.1 and Prop. 6.2].  $\square$

For any  $2 < p \leq \infty$  it is known that there exists a Banach space without  $p$ -AP [SK02, Thm. 6.7], and therefore by Proposition 3.2 there exists an operator space without  $p$ -OAP: simply endow the aforementioned Banach space example with any compatible operator space structure. It would be interesting to find an operator space that has the  $p$ -AP but fails the  $p$ -OAP, although such

Such a  $V$  satisfies the  $\tau_\infty$  density condition by the first paragraph, but fails  $p$ -OAP by construction.

For  $1 \leq p \leq 2$ , the source paper proves that every operator space has  $p$ -OAP. Hence no example failing  $p$ -OAP can exist in that range.  $\square$

## Verification notes

The proof uses only the definitions of complete compactness,  $\tau_\infty$ , and finite-rank completely bounded maps. The key point is that the finite-dimensional projection  $P : W \rightarrow E$  need not have any uniform bound as the completely compact test set and  $\varepsilon$  vary. This is why the argument does not prove norm-density or bounded approximation.

There is an important collateral implication for reviewers. The same lemma shows that condition (iv) in Theorem 3.8 of the source paper, if read literally as unbounded  $\tau_{cc}$ -density, is automatic for all operator spaces  $V$ . Since the paper also states that spaces failing  $p$ -OAP exist for  $p > 2$ , Theorem 3.8(iv) appears to require a correction or an additional boundedness/norm-type hypothesis. This packet is therefore best read both as an answer to the printed open question and as a prompt to check the exact intended topology/density convention.

## Novelty check

On 2026-06-14, I checked the run lightweight indexes for arXiv:2410.05014 and the terms  $p$ -OAP,  $p$ -operator approximation property,  $\tau_\infty$ , operator  $p$ -compactness, and density. I also searched the web for the exact open-question phrase and close variants involving “p-Operator Approximation Property”, “tau\_infty”, “tau\_cc”, and “density condition”. I found no separate answer. The current arXiv page lists v2 as revised on 2025-06-06.

## Human-review recommendation

Verify the intended meaning of the source phrase “density condition in  $\tau_\infty$ ”. If density is meant in the ordinary locally convex topological sense, with no boundedness condition on the approximating finite-rank maps, the answer above is complete. If the authors intended bounded nets or norm/ideal-norm density, the printed question should be reinterpreted and this packet should be reclassified as a wording/proof-gap note rather than a full solution.

## References

- [1] J. A. Chavez-Dominguez, V. Dimant, and D. Galicer, *The  $p$ -Operator Approximation Property*, arXiv:2410.05014, 2024; v2 revised 2025.

- [2] D. P. Sinha and A. K. Karn, *Compact operators whose adjoints factor through subspaces of  $\ell_p$* , Studia Math. **150** (2002), no. 1, 17–33.